

CSCI 6371 Robotics Assignment 2 (100 pts)

Submission:

1. Submit a single Python file (**LastName_studentID_hw2.py**) through Blackboard learn.
2. Do not compress your files.
3. If there is any plagiarism, you will lose all points on the questions at first time. In next, you will lose all points in the whole homework.
4. You can submit your homework **3 times** before the deadline.

Questions: Rotation matrices

Rotation on the X axis

$$R_X = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

Rotation on the Y axis

$$R_Y = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

Rotation on the Z axis

$$R_Z = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Any rotation could be decomposed into the sequence of three elementary matrices. For example, the most intuitive is that which is obtained first by performing a rotation on the X axis by an angle φ (phi), then on the Y axis by an angle θ (theta) and finally on the Z axis by an angle ψ (psi), $R_X(\varphi) \rightarrow R_Y(\theta) \rightarrow R_Z(\psi)$.

For the derivation of rotation matrices, please refer to the attached RotationDerivation.pdf.

1. Please implement the rotation matrices in Python. The skeleton of the code is in HW2_Rotation_matrices_skeleton.py and it is provided in the assignment. (40 pts)
2. Given $\varphi = \pi/2$; $\theta = \pi/4$; $\psi = \pi/2$ and the vector $v_1 = (1,2,3)$, calculate the new vector v_2 after the XYZ rotation on v_1 . (40 pts)
3. Given $\varphi = \pi/2$; $\theta = \pi/4$; $\psi = \pi/2$ and the vector $v_1 = (1,2,3)$, calculate the new vector v_3 after the YZX rotation on v_1 . (20 pts)